Chance-Constrained Personalized LLM Routing with Online Preference Learning

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Abstract

We study personalized LLM routing, where a router selects among heterogeneous models with different costs and capabilities. The router must minimize cumulative inference cost while ensuring reliable realized response quality, even though user preferences are unknown and must be learned from feedback during deployment. We formulate this problem as a constrained contextual bandit and develop a primal-dual GLM-UCB algorithm that learns preferences while balancing cost and quality. Under a logistic preference model and a strict-feasibility condition, we establish sublinear expected cost regret relative to an optimal stationary randomized contextual oracle and a high-probability guarantee on realized cumulative satisfaction. The framework extends beyond LLM routing to broader online model-selection problems with reliability requirements.

1. Introduction

Modern machine learning systems increasingly rely on collections of models with different capabilities and inference costs. Rather than using the same model for every input, a system can route each request to an appropriate model, using inexpensive models when they are sufficient and reserving more expensive models for difficult inputs. This model-routing paradigm is particularly relevant for large language models (LLMs), whose costs and capabilities can differ substantially across models (Chen et al., 2024a; Hu et al., 2024; Ong et al., 2025).

The challenge becomes more subtle when the appropriate model depends not only on the request but also on the user. Different users may prefer different models for the same request, and these preferences may not be known before deployment. A router must therefore learn a user's preferences from feedback while making routing decisions, gradually discovering which model is more likely to satisfy that user for a given request. This connection between personalization and online learning has long been studied through contextual-bandit methods for personalized recommendation (Li et al., 2010; Xu et al., 2020), and recent work has begun to explore personalized and online LLM routing (Dai et al., 2025; Xie et al., 2025; Zeng et al., 2026). These developments motivate an online routing problem in which user preferences are unknown and must be learned from the outcomes of the models selected by the router.

Learning user preferences, however, is only part of the challenge. In many deployments, consistently poor responses may be unacceptable even if average satisfaction is high. A router may instead need to ensure that at least a $(1-\epsilon)$ fraction of responses satisfy a desired quality criterion over a horizon of T requests, with probability at least $1-\delta$. If $Y_t \in \{0,1\}$ indicates whether the response at round t satisfies the user's requirements, this requirement is

$$\mathbb{P}\left(\sum_{t=1}^{T} Y_t \geq (1-\epsilon)T\right) \geq 1-\delta. \tag{1}$$

Here, the satisfaction outcome depends on both the selected model and the user's preferences. Thus, the router must learn the preferences that determine the likelihood of satisfaction while also ensuring that the realized sequence of outcomes remains sufficiently reliable.

This leads to a challenging online decision problem. The router must learn from the limited feedback generated by its own model choices, keep cumulative inference cost low, and ensure that the realized number of satisfactory responses does not fall too far below the target. This differs from an expectation-level constraint, which can balance expected cost and expected satisfaction through a Lagrangian objective. Here, controlling expected satisfaction alone does not directly control the probability of falling below the desired level in the realized sequence. The router must therefore account for both uncertainty in what it has learned about the user's preferences and randomness in the resulting satisfaction outcomes.

We formulate this setting as a constrained contextual bandit with known model costs and unknown preference parameters. At each round, the learner observes the request context, selects one model, incurs its cost, and observes only the satisfaction feedback from that model. The goal is to minimize cumulative cost while satisfying the high-probability

requirement in (1).

Our solution follows a simple principle: learn what the user prefers while continuously monitoring whether the routing decisions are keeping pace with the desired satisfaction level. The router maintains optimistic estimates of how likely each model is to satisfy the user, allowing it to explore uncertain choices while favoring cost-effective ones. At the same time, it keeps track of any accumulated deficit in satisfaction. When the router begins to fall behind, it shifts toward models that are more likely to satisfy the user. We use this idea to propose an algorithm and provide sublinear regret bounds with respect to a stationary routing policy while ensuring that the quality constraints are met.

Although personalized LLM routing motivates the problem, the underlying framework is broader. It applies whenever a system repeatedly chooses among heterogeneous models, services, or alternatives with different costs, while learning an unknown preference or satisfaction model from feedback and seeking to maintain a reliability requirement. This includes, for example, personalized recommendation and online advertising under resource or budget constraints (Zhao & Yang, 2024), cloud or edge service selection subject to cost, quality-of-service, and reliability requirements (Ma et al., 2026), and dynamic procurement among suppliers with heterogeneous costs and uncertain quality under operational constraints (Vassos et al., 2023).

Taken together, these considerations lead to the following contributions:

- We formulate personalized online LLM routing as a contextual bandit with unknown user preferences and a cumulative requirement on realized satisfaction, going beyond the usual expectation-level cost-quality formulation.
- We develop a primal-dual GLM-UCB algorithm that combines optimistic preference learning with a virtual satisfaction-deficit queue and a safety buffer.
- Under a logistic preference model and a uniform strict-feasibility condition, we establish sublinear expected cost regret relative to an optimal stationary randomized contextual oracle while guaranteeing the required realized cumulative satisfaction with high probability.
- Finally, we provide numerical experiments to demonstrate the effectiveness of our algorithm. We evaluate its performance through synthetic experiments and a real-world LLM routing case study, examining constraint satisfaction, routing cost, personalization, and finite-horizon behavior.

2. Related Work

LLM routing and personalization. LLM routing reduces inference costs by selecting among models with different capabilities and prices. Early approaches, such as FrugalGPT (Chen et al., 2024a), use cascades to balance response quality and cost, while subsequent methods learn routing policies from offline model evaluations, preferences, or task information. Examples include RouteLLM (Ong et al., 2025), RouterDC (Chen et al., 2024b), and CARROT (Somerstep et al., 2025). Benchmarks such as RouterBench (Hu et al., 2024) support systematic evaluation of these approaches. However, conventional routing methods primarily focus on learning cost-effective model selection from available data, rather than jointly learning user preferences online and enforcing a reliability constraint on realized outcomes.

Personalized routing addresses the additional challenge that different users may prefer different models. PersonalizedRouter (Dai et al., 2025) models heterogeneous user preferences, while GMTRouter (Xie et al., 2025) and MetaRouter (Zeng et al., 2026) explore learning-based approaches to personalized routing. BaRP (Wei et al., 2025a) further studies routing under bandit feedback. These methods bring routing closer to our setting, in which user preferences must be learned through interactions rather than assumed to be known in advance. However, our work differs from these approaches by combining online personalization with an explicit high-probability guarantee on realized cumulative satisfaction.

Online learning with constraints. Primal–dual methods provide a general framework for online learning under cumulative constraints. Prior work studies long-term constraints in online convex optimization (Mahdavi et al., 2012), constrained Markov decision processes (Cardoso et al., 2019), and contextual bandits with knapsacks (Agrawal & Devanur, 2016; Badanidiyuru et al., 2018; Balseiro et al., 2020). These methods typically regulate resource consumption or optimize expected performance under cumulative constraints. In contrast, our method uses a virtual queue to track the *realized satisfaction deficit* and learns unknown satisfaction probabilities from contextual bandit feedback. This enables a high-probability guarantee on realized cumulative satisfaction, rather than merely controlling expected performance or constraint violation.

Table 1 summarizes how representative routing approaches compare along three dimensions: online learning, personalization, and explicit satisfaction or safety guarantees. Among the approaches considered, our method is the only one that combines all three.

Table 1. Comparison of LLM routing approaches.

Method	Online	Personalized	Safety
FrugalGPT	×	×	×
RouteLLM	×	×	×
PersonalizedRouter	×	✓	×
MetaRouter	✓	✓	×
BaRP	✓	✓	×
Conformal Routing	×	×	✓
SLARouter	✓	×	✓
Ours	✓	✓	✓

References: FrugalGPT (Chen et al., 2024a); RouteLLM (Ong et al., 2025); PersonalizedRouter (Dai et al., 2025); MetaRouter (Zeng et al., 2026); BaRP (Wei et al., 2025a); Conformal Routing (Uddin & Bauer, 2026); SLARouter (Woisetschläger et al., 2026).

3. Problem Formulation

We study personalized LLM routing in which a router selects one of K candidate models for each incoming prompt. The models may differ in computational cost, and a user's preferred model may depend on the prompt. Since these preferences are unknown, the router must learn them from user feedback while keeping inference cost low and maintaining a desired level of satisfaction.

We formulate this problem as a contextual bandit with generalized linear feedback and a cumulative chance constraint. The router observes the prompt and features of all candidate models, but receives satisfaction feedback only from the selected model.

3.1. Online Routing Model

Consider a routing system operating over a horizon of T rounds, indexed by $t \in [T] := \{1, \dots, T\}$. At each round:

1. **Context.** The router observes a prompt $p_t \in \mathcal{P}$, drawn independently from the prompt distribution.
2. **Model selection.** After observing p_t, the router selects
$$a_t \in \mathcal{A} := \{1, \dots, K\}.$$
3. **Features.** A known feature map ϕ assigns a feature vector to each prompt-model pair:
$$x_{t,a} := \phi(p_t, a) \in \mathbb{R}^d, \qquad a \in \mathcal{A}. \tag{2}$$
Thus, the router can compute the features of all candidate models before making its selection.
4. **Cost and feedback.** Selecting model a_t incurs a known deterministic cost
$$c(a_t) \in [0, 1]. \tag{3}$$
After the response is generated, the router observes binary satisfaction feedback $Y_t \in \{0, 1\}$, where $Y_t = 1$ indicates that the response satisfies the user's requirements.

The router therefore observes the features of every candidate model but only observes satisfaction feedback for the selected model.

3.2. User Satisfaction Model

We model the probability that a response satisfies the user through a generalized linear model with a logistic link:

$$Y_t \sim \text{Bernoulli}(p^*_{a_t,t}), \tag{4}$$

where

$$p^*_{a,t} := \mathbb{P}(Y_t = 1 \mid x_{t,a}) = \mu(x_{t,a}^\top \theta^*), \tag{5}$$

and

$$\mu(z) \equiv \frac{1}{1 + e^{-z}}. \tag{6}$$

Here, $\theta^* \in \mathbb{R}^d$ is unknown and captures how the observed prompt-model features relate to the user's satisfaction.

Assumption 3.1 (Boundedness)**.** For every $t \in [T]$ and $a \in \mathcal{A}$,

$$\|x_{t,a}\|_2 \le L, \qquad \|\theta^*\|_2 \le S, \tag{7}$$

where $L, S > 0$ are known. Moreover, the model costs are normalized so that they satisfy $c(a) \in [0, 1]$ for all $a \in \mathcal{A}$.

To ensure that the cumulative satisfaction requirement remains feasible, we assume the availability of a reliable fallback model with known satisfaction performance.

Assumption 3.2 (Fallback)**.** There exists a baseline model $\tilde{a} \in \mathcal{A}$ that is exempt from the generalized linear model and has a known satisfaction probability $p^*_{\tilde{a},t} = \nu$, $\forall t \in [T]$. We additionally assume $\nu > 1 - \epsilon$, so the fallback is strictly feasible; its feedback is not used to estimate θ^*.

The strict-feasibility assumption provides a sufficient condition for feasibility and facilitates our theoretical analysis by ensuring that a baseline action satisfies the target with a positive margin. This assumption does not imply that every LLM routing problem necessarily admits such an action. In practice, the fallback may correspond to a more expensive model that provides sufficiently reliable responses to user queries or a validated service with a known satisfaction guarantee exceeding the target.

We use the following slack factor to quantify this margin.

Definition 3.3 (Slack Factor)**.** For $\epsilon > 0$ and fallback satisfaction probability ν, define the slack factor ξ by

$$\xi := \frac{1 - \nu}{\epsilon}. \tag{8}$$

The strict-feasibility assumption $\nu > 1 - \epsilon$ ensures $0 \le \xi < 1$.

3.3. Chance-Constrained Routing Objective

The router seeks to minimize cumulative inference cost while maintaining a prescribed level of cumulative user satisfaction. Rather than constraining only the expected satisfaction, we require the realized satisfaction over the entire horizon to meet the target with high probability.

Definition 3.4 (Satisfaction Constraint)**.** For a target level $1-\epsilon \in (1/2, 1)$, failure probability $\delta \in (0, 1/2)$, and a known time horizon T, the routing policy must satisfy

$$\mathbb{P}\left(\frac{1}{T}\sum_{t=1}^{T} Y_t \geq 1-\epsilon\right) \geq 1-\delta. \tag{9}$$

Equivalently, at least $(1-\epsilon)T$ responses must satisfy the user with probability at least $1-\delta$. This constraint directly captures the reliability requirement of the routing system.

3.4. Stationary Contextual Oracle

To measure the cost of online learning, we compare the router with a stationary randomized contextual policy that knows the true satisfaction model.

Definition 3.5 (Stationary Contextual Oracle)**.** Let Π denote the class of stationary randomized contextual policies

$$\pi : \mathcal{P} \to \Delta(\mathcal{A}),$$

where $\pi(a \mid p)$ is the probability of selecting model a for prompt p. The oracle is any solution to

$$\pi^* \in \arg\min_{\pi\in\Pi} \mathbb{E}\left[\sum_{t=1}^{T} c(A_t)\right] \tag{10}$$

subject to

$$\mathbb{P}\left(\frac{1}{T}\sum_{t=1}^{T} Y_t(A_t) \geq 1-\epsilon\right) \geq 1-\delta, \tag{11}$$

where $A_t \sim \pi^*(\cdot \mid p_t)$, and the probability is taken over the prompts, model selections, and satisfaction outcomes.

The oracle represents the minimum expected cost that could be achieved by a stationary router with complete knowledge of the user's satisfaction model.

Lemma 3.6 (Necessary Expectation Condition)**.** *Under the stationary contextual oracle, for $\epsilon \in (0.01, 0.5)$ and $\delta \in (0, 0.49)$, there exists a maximum margin $\rho^*_{\max}(T, \epsilon, \delta) = \mathcal{O}(1/\sqrt{T})$ such that*

$$\mathbb{P}\left(\frac{1}{T}\sum_{t=1}^{T} Y_t \geq 1-\epsilon\right) \geq 1-\delta \implies \mathbb{E}[p^*_{a^*_t,t}] \geq 1-\epsilon+\rho^*$$

for every $0 \leq \rho^* \leq \rho^*_{\max}(T, \epsilon, \delta)$.

where Φ denotes the cumulative distribution function (CDF) of the standard normal distribution, and Φ^{-1} denotes its inverse (quantile) function.

Remark 3.7. This lemma shows that, to meet the satisfaction requirement with high probability, the oracle's expected satisfaction must be slightly above the target $1-\epsilon$. Further, the corollary shows that this required gap decreases as $O(1/\sqrt{T})$ and becomes smaller as the horizon T grows.

3.5. Cost Regret

We measure the additional cost incurred while learning relative to the oracle.

Definition 3.8 (Cumulative Cost Regret)**.** Let a^*_t denote the model selected by the oracle at round t. The expected cumulative cost regret is

$$\mathrm{Regret}_T = \mathbb{E}\left[\sum_{t=1}^{T}\big(c(a_t) - c(a^*_t)\big)\right]. \tag{12}$$

Our goal is therefore to achieve sublinear cost regret while satisfying the cumulative satisfaction constraint in (9).

4. Primal-Dual GLM-UCB Router

To devise an online learning algorithm for the problem described in the preceding section, we note that the router must address multiple challenges simultaneously. It must learn which models are likely to satisfy the user while preventing cumulative satisfaction from falling below its target and keeping the cost as low as possible.

We now propose our main algorithm that achieves sublinear regret while satisfying the quality constraint (1). The algorithm proceeds in the following steps:

1. At each round, the router first uses the observed feedback from previous rounds to estimate the unknown GLM parameter θ^*.

2. Next, it selects a model by balancing its inference cost against its estimated satisfaction, with the tradeoff determined by the current dual variable.

3. After observing the realized satisfaction, the router updates the dual variable according to the resulting satisfaction deficit.

The overall algorithm is summarized in Algorithm 1.

Algorithm 1 Primal-Dual GLM-UCB Router

Require: Time horizon T; action set $\mathcal{A}$; parameters ϵ, δ; regularization parameter β; stepsize η; safety buffer ρ.

Initialize $V_1 \leftarrow \beta I_d,\ \lambda_1 \leftarrow 0$.
for $t = 1, \dots, T$ **do**
  Observe prompt p_t and compute $\{x_{a,t}\}_{a\in\mathcal{A}}$.
  Compute $\widehat{\theta}_t$ according to (13).
  for all $a \in \mathcal{A}$ **do**
    Compute $\mathrm{UCB}_t(a)$ according to (16) with confidence radius α_t given by (17).
  end for
  Select

$$a_t \leftarrow \arg\min_{a\in\mathcal{A}} \left[c(a) - \lambda_t \,\mathrm{UCB}_t(a)\right].$$

  Route p_t using model a_t and observe Y_t.
  if $a_t \neq \tilde{a}$

$$V_{t+1} \leftarrow V_t + x_{a_t,t} x_{a_t,t}^\top.$$

  end if
  Update

$$\lambda_{t+1} \leftarrow \max\left\{0,\ \lambda_t + \eta(1 - \epsilon + \rho - Y_t)\right\}.$$

end for

4.1. Learning User Satisfaction

The router estimates θ^* using all feedback observed so far. At round t, it computes the constrained regularized logistic estimator

$$\widehat{\theta}_t = \arg\min_{\theta\in\mathcal{B}_2(S)} \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} l\left(Y_\tau, \mu(x_{a_\tau,\tau}^\top \theta)\right) + \frac{\beta}{2}\|\theta\|_2^2, \tag{13}$$

where $l(y,p) := -y\log p - (1-y)\log(1-p)$ and

$$\mathcal{B}_2(S) = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \le S\}. \tag{14}$$

After observing the selected model's feedback, the design matrix is updated if $a_t \neq \tilde{a}$ according to

$$V_{t+1} = V_t + x_{a_t,t} x_{a_t,t}^\top. \tag{15}$$

The parameter constraint ensures that both the true and estimated parameters lie in the same bounded domain. Since the logistic link has positive derivative on any bounded interval, this gives the uniform curvature needed for the confidence analysis.

4.2. Optimistic Satisfaction Estimates

For every candidate model, we form

$$\mathrm{UCB}_t(a) = \min\{1, \mu(x_{a,t}^\top \widehat{\theta}_t) + \alpha_t \|x_{a,t}\|_{V_t^{-1}}\}. \tag{16}$$

Here, α_t is the confidence radius defined by

$$\alpha_t = \frac{1}{\kappa}\left[\frac{1}{2}\sqrt{d\log\left(1 + \frac{tL^2}{d\beta}\right) + 2\log\left(\frac{2}{\delta_{\mathcal{E}}}\right)} + \sqrt{\beta}S\right], \tag{17}$$

where κ is a constant defined by,[1]

$$\kappa := \inf_{|z|\le LS} \mu'(z) > 0. \tag{18}$$

$$\delta_{\mathcal{E}} := \frac{1}{T^\zeta},\ \zeta \in (2.5, \infty). \tag{19}$$

Moreover, for $a = \tilde{a}$, we set

$$\mathrm{UCB}_t(\tilde{a}) = \nu \tag{20}$$

The confidence-based formula is used for $a \neq \tilde{a}$; the fallback is handled separately by its known value ν. Thus, the router can explore potentially useful models without treating an uncertain model as poor simply because it has not yet received enough feedback.

4.3. Model Selection and Virtual Satisfaction Queue

To balance inference cost against the cumulative satisfaction requirement, we maintain a nonnegative virtual satisfaction queue $\lambda_t \ge 0$, initialized with $\lambda_1 = 0$. The variable λ_t can equivalently be viewed as a dual variable associated with the satisfaction requirement. After selecting model a_t and observing the realized satisfaction Y_t, the queue is updated according to

$$\lambda_{t+1} = \max\left\{0,\ \lambda_t + \eta\left(1 - \epsilon + \rho - Y_t\right)\right\}, \tag{21}$$

where $\eta > 0$ is the step size and $\rho > 0$ is a safety buffer. Thus, the queue increases when the realized satisfaction falls below the buffered target $1 - \epsilon + \rho$ and decreases otherwise.

At round t, given the current queue λ_t, the router selects the model according to

$$a_t = \arg\min_{a\in\mathcal{A}} \left[c(a) - \lambda_t \,\mathrm{UCB}_t(a)\right]. \tag{22}$$

The queue λ_t therefore determines how strongly the router prioritizes satisfaction relative to cost. When λ_t is small, the router places greater emphasis on cost, whereas a larger λ_t favors models with higher optimistic satisfaction estimates.

We choose

$$\eta = \frac{C_\eta}{\sqrt{T}}, \qquad \rho = \frac{C_\rho}{\sqrt{T}} \tag{23}$$

where $C_\eta = \frac{1}{(1-\xi)\epsilon\sqrt{2\log(2/\delta)}}$ and $C_\rho = 4\sqrt{2\log(2/\delta)}$. The safety buffer ρ provides additional slack to account for the random deviation between expected and realized cumulative satisfaction.

[1] We note that under Assumption 3.1, all relevant linear predictors satisfy $|x_{t,a}^\top \theta| \le LS$.

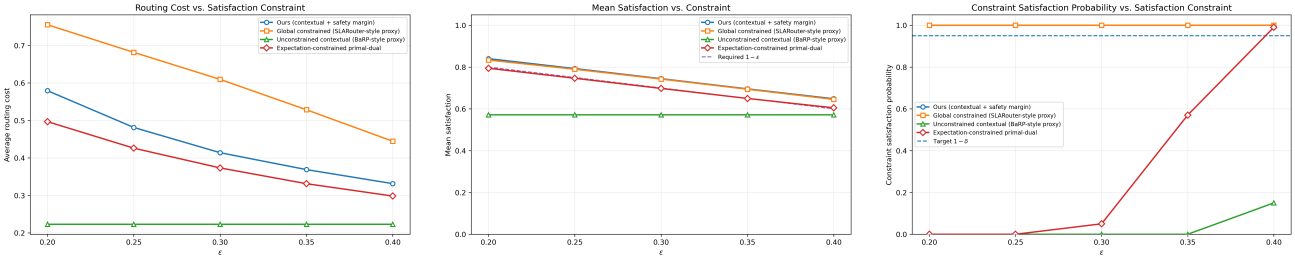


Figure 1. Comparison of the proposed routing algorithm with alternative routing strategies as the satisfaction tolerance ϵ varies. The cost decreases as the constraint is relaxed, while the constraint satisfaction probability is compared against the target $1-\delta$.

5. Theoretical Guarantees

In this section, we establish the theoretical guarantees of our algorithm, including sublinear regret bounds. We first derive a high-probability confidence bound for the estimated satisfaction model, which is then used to control the $\mathrm{UCB}_t(a)$ terms. Building on these results, we analyze the evolution of the virtual queue and establish the main theoretical guarantees of the algorithm.

We begin with the following lemma, which provides a high-probability bound on the estimation error of θ^*.

Lemma 5.1 (Uniform GLM Confidence Bound)**.** *Under Assumption 3.1, with probability at least $1-\delta_{\mathcal{E}}$, simultaneously for all $t \in [T]$ and $a \in \mathcal{A} \setminus \{\tilde{a}\}$,*

$$\|\widehat{\theta}_t - \theta^*\|_{V_t} \le \alpha_t \tag{24}$$

and

$$\left|\mu(x_{a,t}^\top \widehat{\theta}_t) - \mu(x_{a,t}^\top \theta^*)\right| \le \alpha_t \|x_{a,t}\|_{V_t^{-1}}. \tag{25}$$

Next, the following lemma establishes that the router's satisfaction estimate is optimistic, providing an upper bound on the true satisfaction probability. As more feedback is collected, the estimate becomes increasingly accurate, which plays a key role in the subsequent regret analysis.

Lemma 5.2 (UCB Bounds)**.** *On the confidence event of Lemma 5.1, for every $t \in [T]$ and $a \in \mathcal{A} \setminus \{\tilde{a}\}$,*[2]

1. *The UCB is optimistic:*

$$p_{a,t}^* \le \mathrm{UCB}_t(a). \tag{26}$$

2. *The optimism gap satisfies*

$$\mathrm{UCB}_t(a) - p_{a,t}^* \le 2\alpha_t \|x_{a,t}\|_{V_t^{-1}}. \tag{27}$$

5.1. Control of the Satisfaction Queue

Next, we show that the virtual satisfaction queue remains controlled with high probability.

[2]For the fallback action, the UCB is set to its known satisfaction probability ν.

Lemma 5.3 (High-Probability Queue Bound)**.** *Suppose Assumptions 3.1 and 3.2 hold. For any $\delta \in (0,1)$, there exist*

$$B_\lambda = \frac{2}{(1-\xi)\epsilon} + \widetilde{\mathcal{O}}(1/\sqrt{T})$$

such that, with probability at least $1-\delta$,

$$\max_{1 \le t \le T+1} \lambda_t \le B_\lambda. \tag{28}$$

The queue bound in (28) establishes that the dual queue remains bounded throughout the execution of the algorithm. This result is then used to show that the desired satisfaction constraint is satisfied with high probability.

5.2. Main Guarantee

We can now state the main result.

Theorem 5.4 (Cost Regret and Satisfaction Guarantee)**.** *Suppose Assumptions 3.1 and 3.2 hold, $\xi < 1$ and the constants in Algorithm 1 are chosen as specified in Section 4. Then, there exists a constant T_0 such that for $T \ge T_0$ the proposed router satisfies*

$$\mathbb{P}\left(\sum_{t=1}^{T} Y_t \ge (1-\epsilon)T\right) \ge 1-\delta, \tag{29}$$

and its expected cumulative cost regret relative to the optimal stationary contextual oracle satisfies

$$\mathrm{Regret}_T \le \widetilde{\mathcal{O}}\left(\frac{(d+\sqrt{\log(2/\delta)})\sqrt{T}}{(1-\xi)\epsilon}\right) \tag{30}$$

Proof Sketch. The proof consists of two main steps. First, Lemmas 5.1 and 5.2 establish the accuracy of the router's optimistic satisfaction estimates, with estimation errors decreasing as more feedback is collected. Under the decision rule in Eq. (22), the router balances cost against these estimates, allowing the cost regret to be bounded in terms of the estimation errors weighted by the virtual queue λ_t. Lemma 5.3 plays a crucial role by establishing a bound on

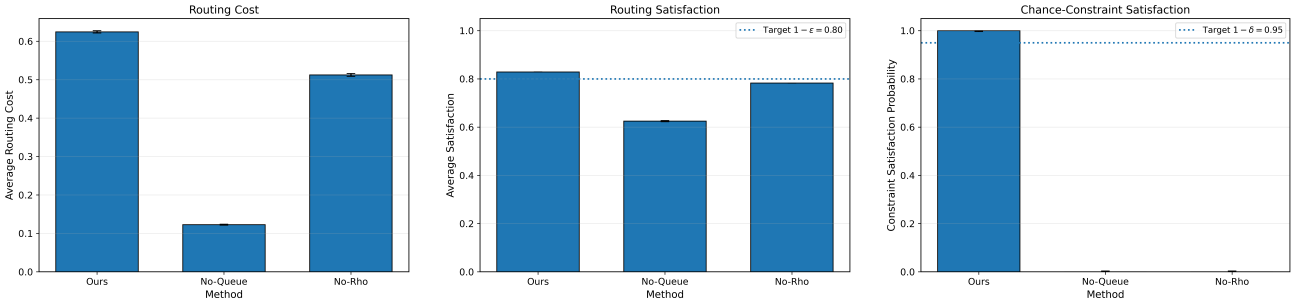


Figure 2. Overall performance and ablation results. Error bars denote 95% confidence intervals. The dotted lines indicate the required satisfaction level $1-\epsilon=0.8$ and constraint probability $1-\delta=0.95$, respectively.

the queue, thereby limiting the impact of estimation errors on regret. Summing these errors over time yields the regret bound in Eq. (30). Second, the queue update in Eq. (21) tracks accumulated satisfaction deficits and increases the queue when realized satisfaction falls below the buffered target. The boundedness of the queue, together with the safety buffer ρ and a concentration argument, ensures that the cumulative satisfaction requirement in Eq. (29) holds with probability at least $1-\delta$. The detailed proofs are provided in Appendices A.4 and A.5. □

Theorem 5.4 shows that the router can learn user preferences while maintaining the required satisfaction level, incurring only sublinear additional cost relative to an oracle. Moreover, the dependence of our regret bounds on the slack factor ξ indicates that achieving lower routing costs becomes increasingly challenging as the fallback performance ν approaches the target satisfaction level $1-\epsilon$.

6. Experimental Results

We evaluate the proposed routing algorithm through five complementary experiments and an additional case study involving real LLMs. We first compare the proposed method with alternative constrained and unconstrained routing strategies, followed by ablation studies that isolate the contributions of the dual queue and safety margin. We then evaluate the method using real LLMs on WildBench v2, investigate the benefits of personalization for users with heterogeneous preferences, and examine its performance as the decision horizon increases. Collectively, these experiments assess chance-constraint satisfaction, routing efficiency, personalization, and finite-horizon performance, while demonstrating the practical applicability of the proposed approach. Implementation details and additional experimental results are provided in Appendix B.

6.1. Comparison with Alternative Routing Strategies

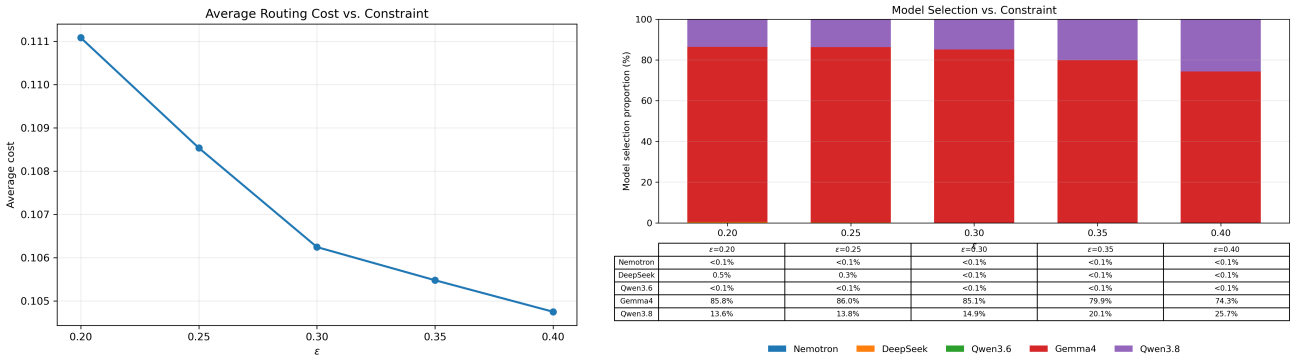
We first compare the proposed method with three representative routing strategies: a SLARouter-style global constrained router (Woisetschläger et al., 2026), a BaRP-style unconstrained personalized router (Wei et al., 2025b), and an expectation-constrained primal-dual router following the standard primal-dual treatment of expectation constraints (Yang et al., 2025). We vary the satisfaction tolerance ϵ while fixing $\delta=0.05$. Hyperparameters are fixed across the ϵ-sweep, except for quantities that necessarily depend on the target constraint, and all methods are evaluated over the same 1000 replay trials.

Figure 1 shows the resulting routing cost, average satisfaction over all replay trials, and empirical constraint satisfaction probability. As ϵ increases, the proposed method reduces its average routing cost because the required satisfaction level $1-\epsilon$ becomes less stringent. At the same time, its empirical constraint satisfaction probability remains close to the target $1-\delta$. We observe that the SLARouter does not exploit contextual information and therefore has a higher average cost, while the BaRP-style router and the expectation-constrained primal-dual router fail to satisfy the quality constraint. These results demonstrate the benefit of combining contextual learning, dual pricing, and the safety margin for chance-constrained routing.

6.2. Ablation

We next examine the contributions of the dual queue and safety margin in the proposed algorithm. We compare **Ours** with **No-Queue**, which removes the λ update equation, and **No-ρ**, which sets the safety margin to $\rho=0$. All methods are evaluated with $\epsilon=0.2$ and $\delta=0.05$. We report average routing cost, average satisfaction, and empirical constraint satisfaction probability with 95% confidence intervals.

As shown in Fig. 2, **Ours** achieves an average satisfaction of 0.829, exceeding the required level $1-\epsilon=0.8$, and satisfies the empirical chance constraint in almost all trials. **No-Queue** and **No-ρ** achieve lower routing costs, but their



	ε=0.20	ε=0.25	ε=0.30	ε=0.35	ε=0.40
Nemotron	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%
DeepSeek	0.5%	0.3%	<0.1%	<0.1%	<0.1%
Qwen3.6	<0.1%	<0.1%	<0.1%	<0.1%	<0.1%
Gemma4	85.8%	86.0%	85.1%	79.9%	74.3%
Qwen3.8	13.6%	13.8%	14.9%	20.1%	25.7%

Figure 3. Real-LLM routing POC: (a) average routing cost as a function of ϵ and (b) model-selection composition as a function of ϵ.

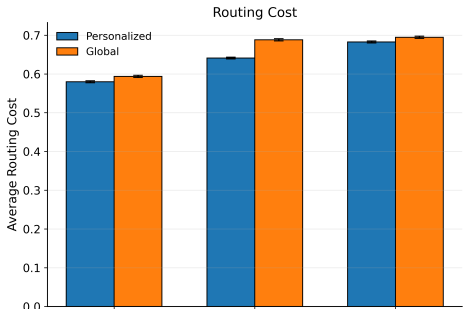


Figure 4. Personalized and global routing costs for Users 1–3.

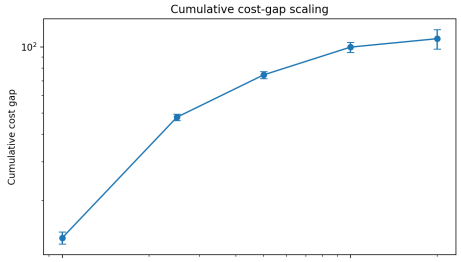


Figure 5. Per-round and cumulative cost gap relative to the oracle across horizons.

empirical constraint-satisfaction probabilities are close to zero. No-ρ also has average satisfaction 0.783, below the target 0.8. These results show that the dual queue is necessary to adapt routing decisions to accumulated constraint violations, while the safety margin provides additional protection against uncertainty in the learned satisfaction estimates.

6.3. Real-LLM Routing

We next evaluate the proposed method on real LLMs using 100 queries from WildBench v2 and five available LLMs as routing actions. For each query–model pair, the response is represented by a six-dimensional quality vector measuring correctness, relevance, completeness, clarity, conciseness, and safety, evaluated by a lightweight judge model (muse-glimmer-30b). We then apply a fixed parameter θ^* in the logistic model of Section 3.2 to generate simulated satisfaction outcomes.

Figure 3 illustrates the routing cost and model selection as the satisfaction tolerance ϵ varies. The average routing cost decreases as ϵ increases, since a lower satisfaction level permits more frequent selection of inexpensive models. In particular, the router increasingly selects Qwen3.8-27B, the lowest-cost model, as the constraint is relaxed. Conversely, for smaller ϵ, it places greater weight on higher-satisfaction models such as DeepSeek-V4-Flash despite their higher cost. This demonstrates that the learned routing policy adapts model selection to the desired quality–cost trade-off induced by the satisfaction constraint.

6.4. Personalization

We next evaluate whether explicitly modeling user-specific preferences can improve routing efficiency. We consider three users who receive the same sequence of requests but have different preference parameters. We compare a personalized learner, which maintains separate model estimates and safety queues for each user, with a global learner that pools observations across users. Both methods use the same requests and counterfactual outcomes.

Figure 4 reports the average routing cost for each user. The personalized learner achieves lower routing cost while maintaining satisfaction above the required level. We omit a separate satisfaction plot because all methods satisfy the constraint in this experiment; therefore, the cost comparison more clearly illustrates the benefit of personalization.

6.5. Finite-Horizon Scaling

Finally, we evaluate how the proposed method behaves as the number of routing decisions increases. We consider horizons $T \in \{100, 250, 500, 1000, 2000\}$ and compare the cumulative cost of the proposed method with that of an

oracle policy that knows the true model parameters.

As shown in Fig. 5, the cumulative cost gap relative to the oracle grows sublinearly over the evaluated horizons. This empirical behavior is consistent with the sublinear regret guarantee established in our analysis. Satisfaction results are omitted for brevity.

7. Conclusion

We presented an online personalized LLM routing framework that learns user preferences while minimizing inference costs and maintaining a high-probability satisfaction constraint. Our primal-dual GLM-UCB algorithm combines optimistic preference learning with a satisfaction-deficit queue and achieves sublinear cost regret while ensuring satisfaction with high probability. Experimental results demonstrate that the proposed method can reduce routing costs through personalization while maintaining the required quality level, including in a real-LLM routing task.

Future research directions include accommodating nonstationary user preferences and incorporating richer feedback signals. Additionally, the current analysis assumes a uniform fallback satisfaction level. Extending the analysis to accommodate context-dependent or uncertain fallback guarantees is another interesting direction for future work.

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A. Proofs

I have made my edits up to this point and will review the proofs later. If the main body of the paper needs to be limited to 8 pages for ICML, you can move the last two plots/experiments to the appendix. Otherwise, the rest looks good to me, and you can send it to Srikant for his feedback.

A.1. Necessary Expectation Condition

Lemma A.1 (Necessary Expectation Condition)**.** *Suppose the stationary contextual oracle generates i.i.d. satisfaction outcomes under the model described in Section 3. In particular, prompts are i.i.d., the oracle randomizes independently across rounds according to $\pi^*(\cdot \mid p_t)$, and satisfaction randomness is conditionally independent across rounds. Given $\epsilon \in (0,1)$, $\delta \in (0,1)$, and T, define*

$$q \equiv 1-\epsilon, \qquad m_T \equiv \lceil Tq \rceil.$$

Let $\mu_{T,\delta} \in (0,1)$ denote the unique value satisfying

$$\mathbb{P}_{\mu_{T,\delta}}\left(S_T \geq m_T\right) = 1-\delta, \tag{31}$$

where

$$S_T \sim \text{Binomial}(T, \mu_{T,\delta}).$$

Define the maximum expectation margin

$$\rho^*_{\max}(T,\epsilon,\delta) \equiv \mu_{T,\delta} - (1-\epsilon). \tag{32}$$

Then, for any

$$0 \leq \rho^* \leq \rho^*_{\max}(T,\epsilon,\delta),$$

satisfying the chance constraint

$$\mathbb{P}\left(\frac{1}{T}\sum_{t=1}^{T} Y_t \geq 1-\epsilon\right) \geq 1-\delta \tag{33}$$

necessarily requires

$$\mathbb{E}\left[p^*_{a^*_t,t}\right] \geq 1-\epsilon+\rho^*. \tag{34}$$

In particular, $\rho^*_{\max}(T,\epsilon,\delta)$ is the largest expectation margin implied by the chance constraint.

Proof. Let

$$\mu^* \equiv \mathbb{E}[p^*_{a^*_t,t}]$$

and define

$$q \equiv 1-\epsilon, \qquad m_T \equiv \lceil Tq \rceil.$$

Under the stated assumptions, the random variables $p^*_{a^*_t,t}$ are i.i.d., and

$$Y_t \mid p^*_{a^*_t,t} \sim \text{Bernoulli}(p^*_{a^*_t,t}).$$

Consequently, the marginal outcomes are i.i.d. Bernoulli with mean μ^*, so that

$$S_T \equiv \sum_{t=1}^{T} Y_t \sim \text{Binomial}(T, \mu^*).$$

The chance constraint can therefore be written as

$$\mathbb{P}_{\mu^*}(S_T \geq m_T) \geq 1 - \delta.$$

Let $\mu_{T,\delta}$ be defined by (31). The binomial upper-tail probability

$$\mathbb{P}_{\mu}(S_T \geq m_T)$$

is continuous and strictly increasing in $\mu \in (0,1)$. Therefore, if

$$\mu^* < \mu_{T,\delta},$$

then

$$\mathbb{P}_{\mu^*}(S_T \geq m_T) < \mathbb{P}_{\mu_{T,\delta}}(S_T \geq m_T) = 1 - \delta.$$

Thus the chance constraint cannot hold when $\mu^* < \mu_{T,\delta}$. Hence, satisfying the chance constraint necessarily requires

$$\mu^* \geq \mu_{T,\delta}.$$

By the definition of $\rho^*_{\max}$,

$$\mu_{T,\delta} = 1 - \epsilon + \rho^*_{\max}(T, \epsilon, \delta).$$

Therefore,

$$\mathbb{E}[p^*_{a^*_t,t}] = \mu^* \geq 1 - \epsilon + \rho^*_{\max}(T, \epsilon, \delta).$$

Since

$$\rho^* \leq \rho^*_{\max}(T, \epsilon, \delta),$$

we obtain

$$\mathbb{E}[p^*_{a^*_t,t}] \geq 1 - \epsilon + \rho^*.$$

This proves the result. □

Corollary A.2 (Large-Horizon Approximation)**.** *For fixed $\epsilon \in (0,1)$ and $\delta \in (0,1)$, as $T \to \infty$, the maximum expectation margin defined in Lemma 3.6 satisfies*

$$\rho^*_{\max}(T, \epsilon, \delta) = \frac{\Phi^{-1}(1-\delta)\sqrt{\epsilon(1-\epsilon)}}{\sqrt{T}} + o\left(\frac{1}{\sqrt{T}}\right), \tag{35}$$

where Φ^{-1} denotes the inverse cumulative distribution function of the standard normal distribution. Thus, for sufficiently large T,

$$\boxed{\rho^*_{\max}(T, \epsilon, \delta) \approx \Phi^{-1}(1-\delta)\sqrt{\frac{\epsilon(1-\epsilon)}{T}}.} \tag{36}$$

Proof. By Lemma 3.6, $\rho^*_{\max}(T, \epsilon, \delta)$ is defined through the binomial threshold $\mu_{T,\delta}$ satisfying

$$\mathbb{P}_{\mu_{T,\delta}}\left(\frac{S_T}{T} \geq 1 - \epsilon\right) = 1 - \delta,$$

where $S_T \sim \text{Binomial}(T, \mu_{T,\delta})$. Write

$$\mu_{T,\delta} = 1 - \epsilon + \rho^*_{\max}(T, \epsilon, \delta).$$

For fixed ϵ and δ, the central limit theorem gives

$$\frac{S_T - T\mu_{T,\delta}}{\sqrt{T\mu_{T,\delta}(1-\mu_{T,\delta})}} \Rightarrow N(0,1).$$

Since $\rho^*_{\max}(T, \epsilon, \delta) \to 0$ as $T \to \infty$, we have

$$\mu_{T,\delta}(1-\mu_{T,\delta}) \to \epsilon(1-\epsilon).$$

Consequently,

$$\frac{\rho^*_{\max}(T, \epsilon, \delta)\sqrt{T}}{\sqrt{\epsilon(1-\epsilon)}} \to \Phi^{-1}(1-\delta),$$

which yields (35). □

A.2. Proof of Regret Bounds

Before proving the regret bounds, we first list and prove the results needed for the main theorem.

A.2.1. Proof of Lemma 5.1

Define

$$\kappa \equiv \inf_{|z| \leq LS} \mu'(z) > 0. \tag{37}$$

For a given confidence parameter $\delta_{\mathcal{E}}$, define

$$\alpha_t = \frac{1}{\kappa}\left[\frac{1}{2}\sqrt{d\log\left(1 + \frac{tL^2}{d\beta}\right) + 2\log\left(\frac{2}{\delta_{\mathcal{E}}}\right)} + \sqrt{\beta}S\right].$$

Let $\mathcal{E}$ denote the event on which the corresponding confidence inequality holds uniformly over all rounds.

Lemma A.3 (Uniform GLM Confidence Bound)**.** *Under Assumption 3.1, with probability at least $1 - \delta_{\mathcal{E}}$, simultaneously for all $t \in [T]$ and $a \in \mathcal{A} \setminus \{\tilde{a}\}$,*

$$\|\widehat{\theta}_t - \theta^*\|_{V_t} \leq \alpha_t \tag{38}$$

and

$$\left|\mu(x_{a,t}^\top \widehat{\theta}_t) - \mu(x_{a,t}^\top \theta^*)\right| \leq \alpha_t \|x_{a,t}\|_{V_t^{-1}}. \tag{39}$$

Proof. Define the regularized negative log-likelihood at round t as

$$\mathcal{L}_t(\theta) = \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} l\left(Y_\tau, \mu(x_{a_\tau,\tau}^\top \theta)\right) + \frac{\beta}{2}\|\theta\|_2^2. \tag{40}$$

The estimator $\widehat{\theta}_t$ is obtained via constrained minimization over the ball $\mathcal{B}_2(S) = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \le S\}$:

$$\widehat{\theta}_t = \arg\min_{\theta \in \mathcal{B}_2(S)} \mathcal{L}_t(\theta). \tag{41}$$

Since $\mathcal{L}_t$ is convex and $\mathcal{B}_2(S)$ is a closed convex set, $\widehat{\theta}_t$ satisfies the first-order optimality condition:

$$\langle \nabla \mathcal{L}_t(\widehat{\theta}_t), \theta - \widehat{\theta}_t \rangle \ge 0 \quad \forall \theta \in \mathcal{B}_2(S). \tag{42}$$

Since $\theta^* \in \mathcal{B}_2(S)$ by Assumption 3.1, setting $\theta = \theta^*$ yields:

$$\langle \nabla \mathcal{L}_t(\widehat{\theta}_t), \widehat{\theta}_t - \theta^* \rangle \le 0. \tag{43}$$

Using the fundamental theorem of calculus along the line segment connecting θ^* and $\widehat{\theta}_t$, we obtain:

$$\nabla \mathcal{L}_t(\widehat{\theta}_t) - \nabla \mathcal{L}_t(\theta^*) = \bar{H}_t(\widehat{\theta}_t - \theta^*), \tag{44}$$

where

$$\bar{H}_t = \int_0^1 \nabla^2 \mathcal{L}_t \left(\theta^* + s\left(\widehat{\theta}_t - \theta^*\right)\right) ds. \tag{45}$$

Taking the inner product with $(\widehat{\theta}_t - \theta^*)$ on both sides:

$$\begin{aligned} \|\widehat{\theta}_t - \theta^*\|^2_{\bar{H}_t} &= \langle \nabla \mathcal{L}_t(\widehat{\theta}_t) - \nabla \mathcal{L}_t(\theta^*), \widehat{\theta}_t - \theta^* \rangle \\ &= \langle \nabla \mathcal{L}_t(\widehat{\theta}_t), \widehat{\theta}_t - \theta^* \rangle - \langle \nabla \mathcal{L}_t(\theta^*), \widehat{\theta}_t - \theta^* \rangle \\ &\le -\langle \nabla \mathcal{L}_t(\theta^*), \widehat{\theta}_t - \theta^* \rangle, \end{aligned} \tag{46}$$

where the final step applies the variational inequality (43).

Evaluating the gradient at θ^* gives:

$$-\nabla \mathcal{L}_t(\theta^*) = \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \eta_\tau x_{a_\tau,\tau} - \beta\theta^*, \tag{47}$$

where $\eta_\tau = Y_\tau - \mu(x_{a_\tau,\tau}^\top \theta^*)$ is a conditionally zero-mean martingale difference.

The Hessian at any point θ is:

$$\nabla^2 \mathcal{L}_t(\theta) = \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \mu'(x_{a_\tau,\tau}^\top \theta) x_{a_\tau,\tau} x_{a_\tau,\tau}^\top + \beta I_d. \tag{48}$$

Because $\theta^*, \widehat{\theta}_t \in \mathcal{B}_2(S)$ and $\mathcal{B}_2(S)$ is convex, any point on the line segment $\theta(s) = \theta^* + s(\widehat{\theta}_t - \theta^*)$ for $s \in [0,1]$ satisfies $\|\theta(s)\|_2 \le S$. Under Assumption 3.1, $\|x_{a_\tau,\tau}\|_2 \le L$, which guarantees:

$$|x_{a_\tau,\tau}^\top \theta(s)| \le LS \quad \forall s \in [0,1]. \tag{49}$$

Consequently, $\mu'(x_{a_\tau,\tau}^\top \theta(s)) \ge \kappa \equiv \inf_{|z| \le LS} \mu'(z) > 0$. Therefore, using the fact that $\kappa \le 1$:

$$\begin{aligned} \bar{H}_t &\succeq \kappa \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} x_{a_\tau,\tau} x_{a_\tau,\tau}^\top + \beta I_d \\ &\succeq \kappa \left(\beta I_d + \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} x_{a_\tau,\tau} x_{a_\tau,\tau}^\top \right) = \kappa V_t. \end{aligned} \tag{50}$$

Applying $\bar{H}_t \succeq \kappa V_t$ to (46) along with the Cauchy-Schwarz inequality in the dual norms gives:

$$\begin{aligned} &\kappa \|\widehat{\theta}_t - \theta^*\|^2_{V_t} \le \|\widehat{\theta}_t - \theta^*\|^2_{\bar{H}_t} \\ &\le \left\langle \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \eta_\tau x_{a_\tau,\tau} - \beta\theta^*, \widehat{\theta}_t - \theta^* \right\rangle \\ &\le \left\| \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \eta_\tau x_{a_\tau,\tau} - \beta\theta^* \right\|_{V_t^{-1}} \|\widehat{\theta}_t - \theta^*\|_{V_t}. \end{aligned} \tag{51}$$

Dividing both sides by $\kappa \|\widehat{\theta}_t - \theta^*\|_{V_t}$ yields:

$$\begin{aligned} \|\widehat{\theta}_t - \theta^*\|_{V_t} &\le \frac{1}{\kappa} \left\| \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \eta_\tau x_{a_\tau,\tau} - \beta\theta^* \right\|_{V_t^{-1}} \\ &\le \frac{1}{\kappa} \left(\left\| \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \eta_\tau x_{a_\tau,\tau} \right\|_{V_t^{-1}} + \sqrt{\beta} S \right). \end{aligned}$$

Since $Y_\tau \in \{0,1\}$, the noise η_τ is conditionally $1/2$-sub-Gaussian. Applying the self-normalized vector martingale inequality, with probability at least $1 - \delta_{\mathcal{E}}/2$, uniformly for all t:

$$\begin{aligned} &\left\| \sum_{\substack{\tau=1 \\ a_\tau \neq \tilde{a}}}^{t-1} \eta_\tau x_{a_\tau,\tau} \right\|_{V_t^{-1}} \\ &\le \frac{1}{2} \sqrt{2 \log\left(\frac{2}{\delta_{\mathcal{E}}}\right) + d \log\left(1 + \frac{(t-1)L^2}{\beta d}\right)}. \end{aligned} \tag{52}$$

Combining these results establishes that $\|\widehat{\theta}_t - \theta^*\|_{V_t} \le \alpha_t$.

Finally, by the Mean Value Theorem on the link function

$\mu(\cdot)$ and Cauchy-Schwarz:

$$\begin{aligned}\left|\mu(x_{a,t}^\top\widehat{\theta}_t) - \mu(x_{a,t}^\top\theta^*)\right| &\le \sup_{|z|\le LS} \mu'(z)\cdot\left|x_{a,t}^\top(\widehat{\theta}_t-\theta^*)\right| \\ &\le L_\mu \|x_{a,t}\|_{V_t^{-1}}\|\widehat{\theta}_t-\theta^*\|_{V_t} \\ &\le \alpha_t\|x_{a,t}\|_{V_t^{-1}},\end{aligned} \tag{53}$$

where $L_\mu = \sup_{|z|\le LS}\mu'(z) \le 1$, completing the proof. □

Next, using this result we bound the UCB terms.

A.3. UCB Bounds

Lemma A.4 (UCB Bounds)**.** *On the event $\mathcal{E}$, for every $a \in \mathcal{A}$:*

1. *The prediction error satisfies*

$$\left|\mu(x_{a,t}^\top\widehat{\theta}_t) - p_{a,t}^*\right| \le \alpha_t\|x_{a,t}\|_{V_t^{-1}}. \tag{54}$$

2. *The UCB is optimistic:*

$$\mathrm{UCB}_t(a) \ge p_{a,t}^*. \tag{55}$$

3. *The optimism gap is bounded by*

$$\mathrm{UCB}_t(a) - p_{a,t}^* \le 2\alpha_t\|x_{a,t}\|_{V_t^{-1}}. \tag{56}$$

Proof. Under the confidence event,

$$\|\widehat{\theta}_t - \theta^*\|_{V_t} \le \alpha_t.$$

By the mean-value theorem and the fact that $\mu'(z) \le 1$,

$$\begin{aligned}\left|\mu(x_{a,t}^\top\widehat{\theta}_t) - \mu(x_{a,t}^\top\theta^*)\right| &\le \left|x_{a,t}^\top(\widehat{\theta}_t-\theta^*)\right| \\ &\le \|x_{a,t}\|_{V_t^{-1}}\|\widehat{\theta}_t-\theta^*\|_{V_t} \\ &\le \alpha_t\|x_{a,t}\|_{V_t^{-1}}.\end{aligned} \tag{57}$$

This proves the prediction error bound.

For optimism, the preceding inequality gives

$$\mu(x_{a,t}^\top\widehat{\theta}_t) + \alpha_t\|x_{a,t}\|_{V_t^{-1}} \ge p_{a,t}^*.$$

Since $p_{a,t}^* \le 1$, taking the minimum with 1 preserves the inequality, and hence

$$\mathrm{UCB}_t(a) \ge p_{a,t}^*.$$

Finally,

$$\begin{aligned}\mathrm{UCB}_t(a) - p_{a,t}^* &\le \left|\mu(x_{a,t}^\top\widehat{\theta}_t) - p_{a,t}^*\right| + \alpha_t\|x_{a,t}\|_{V_t^{-1}} \\ &\le 2\alpha_t\|x_{a,t}\|_{V_t^{-1}}.\end{aligned}$$

□

A.3.1. Bounding the dual variable

Lemma A.5 (Uniform Boundedness of the Dual Variable)**.** *Suppose Assumptions 3.1 and 3.2 hold. For any $\delta \in (0,1)$, there exist $B_\lambda = \frac{2}{(1-\xi)\epsilon} + \widetilde{\mathcal{O}}(1/\sqrt{T})$ such that, with probability at least $1-\delta$,*

$$\lambda_t \le B_\lambda, \qquad \forall t \in [T]. \tag{58}$$

Proof. We first establish a high-probability bound on the dual variable. Define

$$g_t \equiv 1 - \epsilon + \rho - Y_t,$$

and let

$$\mu_t \equiv \mathbb{E}[g_t \mid \mathcal{F}_t], \qquad X_t \equiv g_t - \mu_t.$$

Then

$$\mathbb{E}[X_t \mid \mathcal{F}_t] = 0.$$

Since $Y_t \in \{0,1\}$,

$$|X_t| \le 1.$$

On the confidence event, Assumption 3.2 implies

$$\mathrm{UCB}_t(\tilde{a}) \ge p_{\tilde{a},t}^* = \nu$$

By the primal minimization rule,

$$c(a_t) - \lambda_t\,\mathrm{UCB}_t(a_t) \le c(\tilde{a}) - \lambda_t\,\mathrm{UCB}_t(\tilde{a}).$$

Consequently,

$$\mathrm{UCB}_t(a_t) \ge \mathrm{UCB}_t(\tilde{a}) - \frac{C_{\max}}{\lambda_t},$$

where

$$C_{\max} \equiv \max_{a,a'\in\mathcal{A}} |c(a) - c(a')| \le 1.$$

Therefore,

$$\mathrm{UCB}_t(a_t) \ge \nu - \frac{C_{\max}}{\lambda_t}.$$

Using the confidence bound,

$$p_{a_t,t}^* \ge \mathrm{UCB}_t(a_t) - 2\alpha_t\|x_{a_t,t}\|_{V_t^{-1}},$$

and hence

$$\mu_t = 1 - \epsilon + \rho - p^*_{a_t,t} \tag{59}$$

$$\leq \frac{C_{\max}}{\lambda_t} + 2\alpha_t \|x_{a_t,t}\|_{V_t^{-1}} + \rho - \epsilon + (1-\nu) \tag{60}$$

Now, we consider the case where $\rho + (1-\nu) \leq c_1\epsilon$ where $1 > c_1 \geq \xi$. Therefore, we choose $c_1 = \xi + \rho/\epsilon$. Also, note that $\rho = \frac{C_\rho}{\sqrt{T}}$ and $\xi < 1$. Therefore, for sufficiently large T, $c_1 < 1$.

$$\mu_t \leq \frac{C_{\max}}{\lambda_t} + 2\alpha_t \|x_{a_t,t}\|_{V_t^{-1}} - (1-c_1)\epsilon \tag{61}$$

In particular, whenever

$$\lambda_t \geq \frac{C_{\max}}{\epsilon c_2},$$

for $0 < c_2 < 1 - c_1$ we have

$$\mu_t \leq 2\alpha_t \|x_{a_t,t}\|_{V_t^{-1}} - (1 - c_1 - c_2)\epsilon. \tag{62}$$

Define an exceptional round to be a round satisfying

$$2\alpha_t \|x_{a_t,t}\|_{V_t^{-1}} > (1 - c_1 - c_2)\epsilon$$

On every non-exceptional round for which $\lambda_t \geq C_{\max}/(c_2\epsilon)$, equation (62) gives

$$\mu_t \leq 0.$$

Let

$$\alpha_T \equiv \max_{1 \leq t \leq T} \alpha_t.$$

Every exceptional round satisfies

$$\|x_{a_t,t}\|_{V_t^{-1}} > \frac{(1 - c_1 - c_2)\epsilon}{2\alpha_T}.$$

By the elliptical potential count lemma, the number N_T of exceptional rounds satisfies

$$N_T \leq \frac{2d(2\alpha_T)^2}{(1 - c_1 - c_2)^2\epsilon^2} \log \frac{\det(V_T)}{\det(V_1)} \tag{63}$$

$$\leq \frac{8d\alpha_T^2}{(1 - c_1 - c_2)^2\epsilon^2} \log\left(1 + \frac{TL^2}{\beta d}\right), \tag{64}$$

where we used $V_1 = \beta I$ and $\|x_{a_t,t}\|_2 \leq L$.

We now control excursions of the dual process above the threshold

$$a \equiv \frac{C_{\max}}{c_2\epsilon}.$$

Suppose that $\lambda_t > a$, and let

$$k = \max\{s < t : \lambda_s \leq a\}$$

be the most recent time before t at which the process was at or below a. Since $|g_k| \leq 1$ and $\eta = C_\eta/\sqrt{T}$,

$$\lambda_{k+1} \leq a + \eta.$$

During the excursion, $\lambda_s > a$, and hence the conditional drift is non-positive on all non-exceptional rounds. On an exceptional round, we use the crude bound

$$\mu_s \leq 1.$$

Therefore, writing $\mathcal{I}_{ex}$ for the set of exceptional rounds,

$$\lambda_t \leq a + \eta + \eta \sum_{s=k+1}^{t-1} \mu_s + \eta \sum_{s=k+1}^{t-1} X_s \tag{65}$$

$$\leq a + \eta + \eta|\mathcal{I}_{ex}| + \eta \sum_{s=k+1}^{t-1} X_s \tag{66}$$

$$\leq a + \eta + \eta N_T + \eta \sum_{s=k+1}^{t-1} X_s. \tag{67}$$

Since (X_t) is a martingale-difference sequence with $|X_t| \leq 1$, a maximal Azuma-Hoeffding inequality implies that, with probability at least $1 - \delta'$,

$$\max_{1 \leq k < t \leq T+1} \sum_{s=k}^{t-1} X_s \leq \sqrt{2T \log\left(\frac{2}{\delta'}\right)}.$$

Combining this inequality with (67), we obtain

$$\max_{t \leq T+1} \lambda_t \leq a + \eta + \eta N_T + \eta\sqrt{2T \log\left(\frac{2}{\delta'}\right)}$$

with probability at least $1 - \delta' - \delta_{\mathcal{E}}$.

Using $\eta = C_\eta/\sqrt{T}$ and the bound in (64), it follows that

$$\max_{t \leq T+1} \lambda_t \leq \frac{1}{c_2\epsilon} + \frac{C_\eta}{\sqrt{T}} + C_\eta\sqrt{2\log\left(\frac{2}{\delta'}\right)} + \frac{8dC_\eta\alpha_T^2}{(1 - c_1 - c_2)^2\epsilon^2\sqrt{T}} \log\left(1 + \frac{TL^2}{\beta d}\right) \equiv B_\lambda(T, \delta)$$

with probability at least $1 - \delta' - \delta_{\mathcal{E}}$.

We choose $\delta_{\mathcal{E}} = T^{-\zeta}$ and $\delta' = \delta - \delta_{\mathcal{E}}$, with $\zeta > 2$. Additionally, since other terms are $\tilde{\mathcal{O}}(1/\sqrt{T})$, we can choose $c_2 = (1 - c_1 - T^{-1/8})$. Therefore,

$$B_\lambda = \frac{1}{(1 - c_1 - T^{-1/8})\epsilon} + C_\eta\sqrt{2\log\left(\frac{2}{\delta'}\right)} + \tilde{\mathcal{O}}(1/\sqrt{T}) \tag{68}$$

For sufficiently large T, $\rho \to 0$ and $\delta' \to \delta$, therefore c_1 can be chosen to be ξ. Thus

$$B_\lambda = \mathcal{O}\Big(\frac{1}{(1-\xi)\epsilon} + C_\eta\sqrt{2\log\Big(\frac{2}{\delta}\Big)}\Big) \tag{69}$$

Substituting C_η, we get $B_\lambda = \frac{2}{(1-\xi)\epsilon} + \widetilde{\mathcal{O}}(1/\sqrt{T})$

□

A.4. Proof of the Cost Regret Bound

Proof of Theorem 5.4, regret bound. Let $\mathcal{F}_t \equiv \sigma(p_1, a_1, Y_1, \ldots, p_{t-1}, a_{t-1}, Y_{t-1}, p_t)$, so that a_t (hence $x_{a_t,t}$ and λ_t) is $\mathcal{F}_t$-measurable, while Y_t and $a_t^* \sim \pi^*(\cdot \mid p_t)$ (drawn with independent external randomness) are not. Let also $\mathcal{H}_{t-1} \equiv \sigma(p_1, a_1, Y_1, \ldots, p_{t-1}, a_{t-1}, Y_{t-1})$ denote the history up to the end of round $t-1$. Then $\widehat{\theta}_t$, V_t, α_t and λ_t are $\mathcal{H}_{t-1}$-measurable, while p_t and a_t^* are independent of $\mathcal{H}_{t-1}$.

Fix the failure level $\delta' = 1/T$, used only in this proof, and let $B_\lambda = B_\lambda(T, 1/T)$ be the bound of Lemma 5.3 on $\max_{1\le t\le T+1}\lambda_t$ at this level.

Define

$$\tau \equiv \min\Big\{ t \in \{1,\ldots,T+1\} \;:\; \mathcal{E} \text{ fails at round } t, \text{ or } \lambda_t > B_\lambda \Big\}, \tag{70}$$

with $\tau = T+1$ if neither event occurs. Therefore,

$$\mathbb{P}(\tau \le T) \;\le\; \delta_{\mathcal{E}} + \delta' \;=\; \delta_{\mathcal{E}} + \frac{1}{T}. \tag{71}$$

By definition of τ as the *first* crossing time, $\lambda_t \le B_\lambda$ holds only for $t < \tau$, not at $t = \tau$ itself; we therefore work with $T' \equiv (\tau - 1) \wedge T$ throughout, rather than $\tau \wedge T$.

Step 1: Primal comparison. By the selection rule of Algorithm 1, for every t,

$$c(a_t) - \lambda_t\,\mathrm{UCB}_t(a_t) \;\le\; c(a_t^*) - \lambda_t\,\mathrm{UCB}_t(a_t^*),$$

so

$$c(a_t) - c(a_t^*) \;\le\; \lambda_t\Big(\mathrm{UCB}_t(a_t) - \mathrm{UCB}_t(a_t^*)\Big). \tag{72}$$

Step 2: Replacing UCB by the true satisfaction probabilities. By Lemma 5.2, on $\mathcal{E}$, for every $a \in \mathcal{A}$,

$$p_{a,t}^* \;\le\; \mathrm{UCB}_t(a) \;\le\; p_{a,t}^* + 2\alpha_t\|x_{a,t}\|_{V_t^{-1}}.$$

For $t \le T'$, $\mathcal{E}$ holds at round t by definition of τ, so

$$\mathrm{UCB}_t(a_t) - \mathrm{UCB}_t(a_t^*) \;\le\; \big(p_{a_t,t}^* - p_{a_t^*,t}^*\big) + 2\alpha_t\|x_{a_t,t}\|_{V_t^{-1}}. \tag{73}$$

Combining (72)–(73), for $t \le T'$,

$$c(a_t) - c(a_t^*) \;\le\; \lambda_t\big(p_{a_t,t}^* - p_{a_t^*,t}^*\big) + 2\lambda_t\alpha_t\|x_{a_t,t}\|_{V_t^{-1}}. \tag{74}$$

Step 3: Decomposition. With $g_t \equiv 1 - \epsilon + \rho - Y_t$, $\mu_t \equiv \mathbb{E}[g_t \mid \mathcal{F}_t] = 1 - \epsilon + \rho - p_{a_t,t}^*$, $X_t \equiv g_t - \mu_t = p_{a_t,t}^* - Y_t$, and $\nu_t^* \equiv 1 - \epsilon + \rho - p_{a_t^*,t}^*$, adding and subtracting $1 - \epsilon + \rho$ in (74) gives, for $t \le T'$,

$$c(a_t) - c(a_t^*) \;\le\; \underbrace{-\lambda_t\mu_t}_{(A)_t} + \underbrace{\lambda_t\nu_t^*}_{(B)_t} + \underbrace{2\lambda_t\alpha_t\|x_{a_t,t}\|_{V_t^{-1}}}_{(C)_t}. \tag{75}$$

We bound $\sum_{t=1}^{T'}\mathbb{E}[(A)_t]$, $\sum_{t=1}^{T'}\mathbb{E}[(B)_t]$, and $\sum_{t=1}^{T'}(C)_t$ in Steps 4–6.

Step 4: Term (A). From $\lambda_{t+1} = \max\{0, \lambda_t + \eta g_t\}$ and $|g_t| \le 1$,

$$-\lambda_t g_t \;\le\; \frac{\lambda_t^2 - \lambda_{t+1}^2}{2\eta} + \frac{\eta}{2} \qquad \text{(pathwise)}. \tag{76}$$

Since $(A)_t = -\lambda_t g_t + \lambda_t X_t$, summing (76) over $t = 1, \ldots, T'$ and using $\lambda_1 = 0$,

$$\sum_{t=1}^{T'}(A)_t \;\le\; \frac{\eta T}{2} + \sum_{t=1}^{T'}\lambda_t X_t. \tag{77}$$

Since λ_t is $\mathcal{F}_t$-measurable and $|\lambda_t X_t| \le B_\lambda$ for $t \le T'$, $\big(\sum_{t=1}^{n\wedge(\tau-1)}\lambda_t X_t\big)_{n\ge 0}$ is a bounded martingale, so by optional stopping,

$$\mathbb{E}\Big[\sum_{t=1}^{T'}\lambda_t X_t\Big] = 0 \quad\Longrightarrow\quad \mathbb{E}\Big[\sum_{t=1}^{T'}(A)_t\Big] \;\le\; \frac{\eta T}{2}. \tag{78}$$

Step 5: Term (B). By the proof of Lemma 3.6, feasibility of π^* implies $\mu^* \equiv \mathbb{E}[p_{a_t^*,t}^*] \ge \mu_{T,\delta} = 1 - \epsilon + \rho_{\max}^*$. Because p_t and a_t^* are independent of $\mathcal{H}_{t-1}$ (prompts are i.i.d. and π^* is stationary with external randomization), $\mathbb{E}[p_{a_t^*,t}^* \mid \mathcal{H}_{t-1}] = \mu^*$. Hence

$$\begin{aligned}\mathbb{E}\big[1 - \epsilon + \rho - p_{a_t^*,t}^* \,\big|\, \mathcal{H}_{t-1}\big] &= \rho - (\mu^* - (1-\epsilon)) \\ &\le \rho - \rho_{\max}^* \\ &\le (\rho - \rho_{\max}^*)^+. \end{aligned} \tag{79}$$

Since $\mathbf{1}\{t \le T'\}\lambda_t$ is $\mathcal{H}_{t-1}$-measurable with $0 \le \lambda_t \le B_\lambda$ on $\{t \le T'\}$,

$$\begin{aligned}&\mathbb{E}\Big[\sum_{t=1}^{T'}(B)_t\Big] \\ &= \sum_{t=1}^{T}\mathbb{E}\Big[\mathbf{1}\{t \le T'\}\lambda_t\,\mathbb{E}\Big[1 - \epsilon + \rho - p_{a_t^*,t}^* \mid \mathcal{H}_{t-1}\Big]\Big] \\ &\le T\,B_\lambda\,(\rho - \rho_{\max}^*)^+ \end{aligned} \tag{80}$$

(If $\rho \le \rho_{\max}^*$ this is 0)

Step 6: Term (C). For $t \le T'$, $\lambda_t \le B_\lambda$ and $\alpha_t \le \alpha_T$. By Cauchy–Schwarz and the elliptical potential lemma (with $\beta \ge L^2$),

$$\sum_{t=1}^{T'} (C)_t \;\le\; 2B_\lambda \alpha_T \sqrt{2dT \log\Big(1 + \frac{TL^2}{\beta d}\Big)}. \tag{81}$$

Step 7: Cost of the bad event. Rounds $T'+1, \dots, T$ (nonempty only if $\tau \le T$, i.e. the stopping round τ itself plus everything after) each contribute at most 1, so

$$\begin{aligned}\mathbb{E}\Big[\sum_{t=T'+1}^{T} \big(c(a_t) - c(a_t^*)\big)\Big] \;&\le\; T \cdot \mathbb{P}(\tau \le T) \\ &\le\; T\delta_{\mathcal{E}} + 1 \\ &\le\; 2 \end{aligned} \tag{82}$$

using $\delta_{\mathcal{E}} = T^{-\zeta}$, $\zeta \ge 1$.

Step 8: Combining. Summing (75) and splitting at T', then applying (78), (80), (81), (82),

$$\begin{aligned}\mathrm{Regret}_T \;\le\; &\frac{\eta T}{2} + 2B_\lambda \alpha_T \sqrt{2dT \log\Big(1 + \frac{TL^2}{\beta d}\Big)} \\ &+ TB_\lambda(\rho - \rho^*_{\max})^+ + 2. \end{aligned} \tag{83}$$

With $\eta = C_\eta/\sqrt{T}$ and, from the crude (Azuma-based) form of Lemma 5.3 at $\delta' = 1/T$,

$$B_\lambda \;=\; \mathcal{O}(1/\big(\epsilon(1-\xi)\big)), \qquad \alpha_T \;=\; \mathcal{O}\Big(\sqrt{d \log T}\Big)$$

(the $\log(1/\delta') = \log T$ dependence is absorbed into $\widetilde{\mathcal{O}}(\cdot)$). Choose $\rho = C_\rho/\sqrt{T}$. Then $(\rho - \rho^*_{\max})^+ \le \rho$ and

$$TB_\lambda(\rho - \rho^*_{\max})^+ \le C_\rho B_\lambda \sqrt{T} = \widetilde{\mathcal{O}}\Big(\frac{\sqrt{\log(2/\delta)}\sqrt{T}}{(1-\xi)\epsilon}\Big).$$

Therefore (83) gives

$$\begin{aligned}&\mathrm{Regret}_T \\ &\le \frac{C_\eta}{2}\sqrt{T} + \widetilde{\mathcal{O}}\left(\frac{d\sqrt{T}}{(1-\xi)\epsilon}\right) \\ &+ \widetilde{\mathcal{O}}\left(\frac{\sqrt{\log(2/\delta)}\sqrt{T}}{(1-\xi)\epsilon}\right) + 2 \\ &= \widetilde{\mathcal{O}}\left(\frac{(d + \sqrt{\log(2/\delta)} + \sqrt{1/\log(2/\delta)})\sqrt{T}}{(1-\xi)\epsilon}\right) \\ &= \widetilde{\mathcal{O}}\left(\frac{(d + \sqrt{\log(2/\delta)})\sqrt{T}}{(1-\xi)\epsilon}\right)\end{aligned}$$

We can show that for $\delta \le 1$, $C_\eta\sqrt{T}/2 \le C_\rho B_\lambda \sqrt{T}$, therefore we can simplify the expression to get the last term. □

A.5. Proof of High-Probability Quality Constraint Satisfaction

Proof. Consider the dual update

$$\lambda_{t+1} = \max\{0,\; \lambda_t + \eta\,(1 - \epsilon + \rho - Y_t)\}\,.$$

Since $\max\{0, x\} \ge x$,

$$\lambda_{t+1} \ge \lambda_t + \eta(1-\epsilon) + \eta\rho - \eta Y_t.$$

Summing this inequality from $t = 1$ to T and using $\lambda_1 = 0$ gives

$$\begin{aligned}\lambda_{T+1} &\ge \eta \sum_{t=1}^{T} (1 - \epsilon + \rho - Y_t) \\ &= \eta(1-\epsilon+\rho)T - \eta \sum_{t=1}^{T} Y_t. \end{aligned} \tag{84}$$

Rearranging,

$$\sum_{t=1}^{T} Y_t \ge (1-\epsilon+\rho)T - \frac{\lambda_{T+1}}{\eta}. \tag{85}$$

On the event where the dual variable is uniformly bounded,

$$\lambda_{T+1} \le B_\lambda.$$

Therefore,

$$\sum_{t=1}^{T} Y_t \ge (1-\epsilon)T + \rho T - \frac{B_\lambda}{\eta}. \tag{86}$$

Finally, we have

$$\eta = \frac{C_\eta}{\sqrt{T}}, \qquad \rho = \frac{C_\rho}{\sqrt{T}}.$$

Choose

$$C_\rho > \frac{B_\lambda}{C_\eta}.$$

Then

$$\rho T - \frac{B_\lambda}{\eta} = \left(C_\rho - \frac{B_\lambda}{C_\eta}\right)\sqrt{T} > 0,$$

which is sufficient for the satisfaction guarantee.

Using the queue bound above with $\delta' = \delta - \delta_{\mathcal{E}}$ and $\delta_{\mathcal{E}} = T^{-\zeta}$, its leading terms satisfy

$$\frac{B_\lambda}{C_\eta} \le \frac{1}{(1-\xi)\epsilon C_\eta} + \sqrt{2\log\left(\frac{2}{\delta'}\right)} + o(1).$$

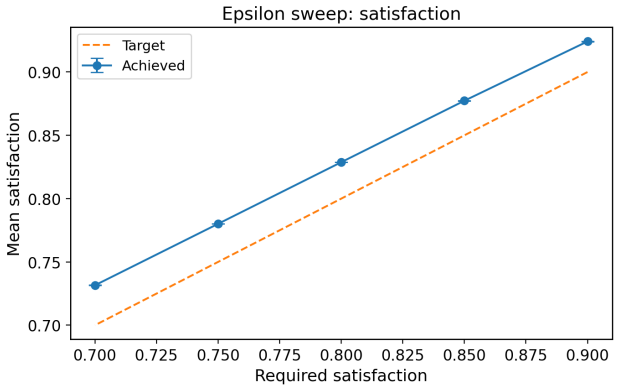


Figure 6. Sensitivity to the satisfaction tolerance ϵ. Dashed lines indicate the corresponding satisfaction and constraint thresholds.

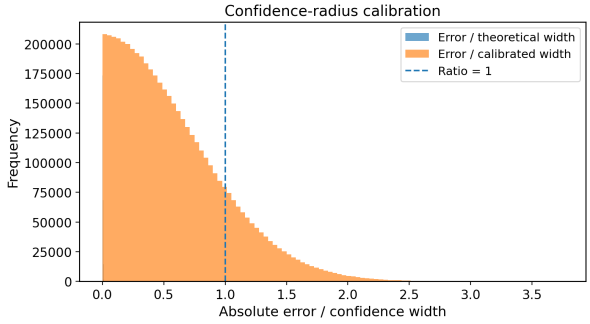


Figure 7. Confidence Calibration

For $C_\eta = ((1-\xi)\epsilon\sqrt{2\log(2/\delta)})^{-1}$, the right-hand side converges to $2\sqrt{2\log(2/\delta)}$ as $T \to \infty$. Since $C_\rho = 4\sqrt{2\log(2/\delta)}$, the strict inequality $C_\rho > B_\lambda/C_\eta$ therefore holds for all sufficiently large T under the stated queue bound. Hence there exists T_0 such that, for $T \geq T_0$, the satisfaction constraint holds with probability at least $1-\delta$.

Consequently,

$$\sum_{t=1}^{T} Y_t \geq (1-\epsilon)T.$$

Since this conclusion holds on an event of probability at least $1-\delta$, we obtain

$$\mathbb{P}\left(\frac{1}{T}\sum_{t=1}^{T} Y_t \geq 1-\epsilon\right) \geq 1-\delta.$$

This establishes that our algorithm satisfies the quality constraint. □

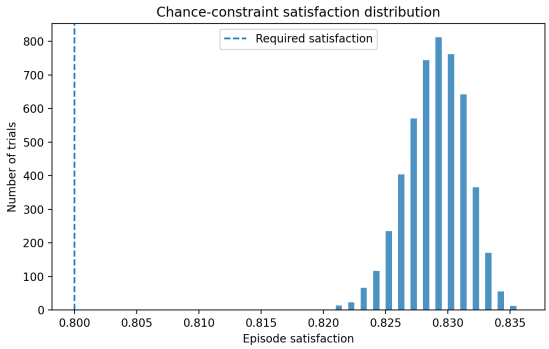


Figure 8. Empirical probability of satisfying the chance constraint over 5000 independent trials.

B. Additional Experiments

B.1. Experimental Details

All experiments are implemented in Python. For the synthetic experiments, we consider feature dimensions $d = 8$ and $K = 5$ routing actions. Unless otherwise stated, we use $T = 1000$ rounds, $\epsilon = 0.2$, $\delta = 0.05$, and initialize the model parameters and covariance matrix as specified in the algorithm. Results are averaged over multiple independent runs, with confidence intervals computed across runs.

For our experiments, for $T = 100$, we fix $\eta = 0.5$, $\rho = 0.15$ for $T = 100$ and further scale them with respect to $1/\sqrt{T}$ for other values of T. Similarly, in the experiments we multiply the theoretical confidence radius α_t by 0.005 as an empirical calibration. This calibrated radius is used only for the experiments. We report their constraint-satisfaction performance empirically.

All experiments were conducted on a CPU-based machine equipped with an Intel Core i5-14600KF processor and 32 GB of RAM. The total runtime for all experiments was approximately six hours.

For the real-LLM experiments, we use 100 queries from WildBench v2 and five LLMs as the available routing actions. The LLMs are mentioned as follows:

1. nemotron-3-super-120b-a12b (NVIDIA, 2025)
2. deepseek-v4-flash (DeepSeek-AI, 2026)
3. qwen3.6-35b-a3b (Qwen Team, 2026a)
4. gemma-4-31b-it (Team, 2026)
5. qwen3.8-27b (Qwen Team, 2026b)

Each response is evaluated by an independent lightweight judge model (muse-glimmer-30b (Labs, 2026)) using six quality dimensions: correctness, relevance, completeness,

clarity, conciseness, and safety. The resulting quality scores are used to compute the satisfaction outcome and routing cost. All methods use the same queries, models, costs, and evaluation procedure to ensure a fair comparison. The code for all the experiments can be found at https://github.com/ameyanjarlekar/LLM-Routing and the models are hosted at https://lumen.ncsa.illinois.edu/.

B.2. Sensitivity to the Satisfaction Tolerance

We study how the routing performance changes with the satisfaction tolerance ϵ. A smaller ϵ imposes a stricter satisfaction requirement, while a larger ϵ allows greater flexibility in model selection. We vary ϵ over a range of values and evaluate the resulting routing cost, satisfaction, and chance-constraint probability.

B.3. Confidence Width Calibration

We evaluate the tightness of the theoretical confidence width for the predicted satisfaction probabilities. At $T = 1000$, the raw theoretical width is highly conservative in our evaluated data. We also evaluate a calibrated width obtained by applying a fixed scaling factor of 0.005. The calibrated width covers 83.4% of the observed errors in this experiment, which justifies our choice of multiplying the theoretical confidence radisu α_t by 0.005.

B.4. Quality-Constraint Satisfaction

We next evaluate the reliability of constraint satisfaction across independent trials. For each independent trial, we run the algorithm for $T = 1000$ rounds and record whether the realized average satisfaction meets the target $1-\epsilon = 0.8$. We repeat this procedure over 5000 independent trials and report the empirical constraint-satisfaction frequency with 95% confidence intervals. Figure 8 compares this empirical frequency with the target $1 - \delta = 0.95$